SASKEN

C-V2X Autonomous Resource Selection Explained

Whitepaper

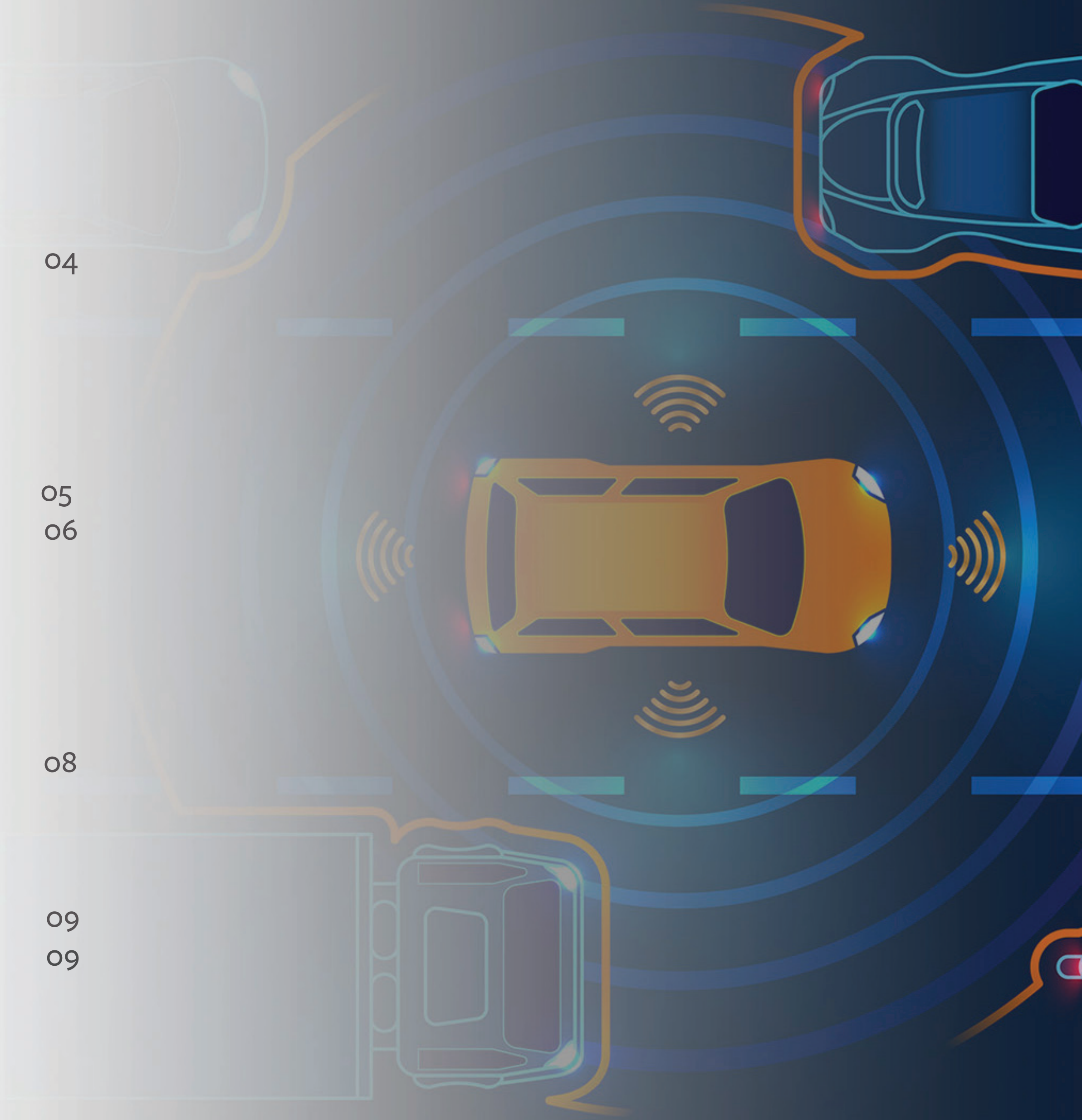
Abstract:

This article explains Autonomous Resource Selection method introduced in 3GPP Release 14, under V2V (Vehicle-to-Vehicle) enhancements. The reader is expected to have basic understanding of LTE air interface protocols and standards.

LTE air interface protocols are based on network presence and network controlled resource allocation. Vehicle to vehicle communication, on the other hand, as the name suggests, does not require network presence and should be possible at all times. A study of 3GPP standards was carried out at Sasken Technologies Ltd. to find out enhancements done to support vehicle to vehicle communication.

Author:
By Samir Amberkar
Principal Engineer, Product Engineering Services,
Sasken Technologies Limited

Table of Content



1 - Background

1.1 - What is Vehicle to Vehicle (V2V)?

With the target of improving public safety (on road) and traffic efficiency, ETSI introduced ITS (Intelligent Transport Systems) standards. The basic idea here is to enable vehicles to communicate or exchange messages/information with each other.

An emergency vehicle may send out a special message to other vehicles on road and upon receiving such message vehicles may alert the driver to the presence of emergency vehicle behind. An emergency vehicle needs to attach its location (and speed/direction) information to this message. Recipient vehicle, along with the knowledge of its own location/speed/direction, need to intelligently deduce whether it is in the path of emergency vehicle and alert the driver accordingly.

Vehicles may exchange their location/speed/direction information with each other with certain minimum periodicity. This then shall help in generating critical alerts to driver like another incoming vehicle at signal-less junction, motorcycle on the curve etc.

With supporting infrastructure nodes, the warning messages can further be relayed to vehicles at longer distances. Some examples are 'broken down vehicle ahead' or 'de-tour ahead' notifications.
ITS has listed out these target scenarios along with expected response from vehicles on detection of those scenarios.

"X" in V2X stand for Infrastructure (RSU) or Pedestrian.

To make V2V possible, RF or wireless access is needed underneath. WLAN (Wi-Fi) was amended (802.11p) to provide support for V2V. 3GPP introduced Proximity Services (ProSe) featured in Release 12 and further enhanced it in Release 14 for V2V.

1.2 - CV2X

3GPP support to V2V is popularly known as Cellular V2X. Additional RF symbols for DeModulation Reference Signals (to support high relative speed) and Enhanced ProSe services are the top level changes done for V2V support in 3GPP context. A new LTE band 47(with 10 and 20 MHz bandwidth) is introduced to support working on (unlicensed) 5.9 GHz spectrum. ProSe has its own set of physical channels with LTE RF chain (OFDM access) and basic physical organization being same.

1.3 - PC5 interface or SideLink

PC5 interface introduced under ProSe services in Rel 12 makes way for UE-2-UE communication. At Layer 2 (Access Stratum) level, PC5 interface boils down to "SideLink" (on the lines on DownLink, UpLink).

1.4 - SideLink enhancements

Below table shows top level view of SideLink enhancements for V2V:

SideLink capabilities	V2V enhancements
Discovery function	Not used
One-to-one and one-to-many communication	Only one-to-many (broadcast) is needed
Transmission mode 1 (Network controlled allocation) and mode 2 (Autonomous selection).	Transmission mode 3 (Network controlled allocation) and mode 4 (Autonomous selection).
Channel (or access) configuration in dedicated SIB Type 18	Channel (or access) configuration in dedicated SIB Type 21
Channel (or access) configuration also in UICC under dedicated service label	Channel (or access) configuration also in UICC under dedicated V2X service label
Out of coverage operation supported	Out of coverage operation supported

1.5 - Latency requirements

Top most requirement that ITS Application layer expect wireless access (Modem) to fulfill is **Latency.** Latency expected from protocol stack (from the time it receives V2V message from Application layer till it is on Air) is 20, 50, or 100 milliseconds (based on type of message). In addition, as V2V messages are periodically transmitted, it is expected that periodicity (messages per second) is maintained.

2 - Problem statement

If we have certain number of **periodic RF** resources (a frequency-time grid) in network control free environment, how can **vehicles in the vicinity** choose resource for transmission of their V2X message **autonomously without direct one-to-one communication** and satisfying their individual **latency**, **periodicity**, and **message size** requirements at the same time?

3 - Solution

Let us split the problem and solution step by step.

3.1 - Resources

In LTE, spectrum is divided in bandwidths, which is further divided in Physical Resource Blocks, each PRB contain 12 consecutive (OFDM) subcarriers, subcarriers are separated by 15 kHz. Time is divided in 10240 millisecond Hyperframes, each hyperframe contain 1024 Frames, and each frame contain 10 Subframes.

Sidelink further organizes PRBs in Subchannels, each subchannel contain certain (as per access configuration) number of PRBs (say 5). In time domain, a Subframe is used as a unit of allocation. Subframe numbers start with zero and resets after 10240 subframes. Access configuration indicate the subframes that may be used for reception (RX) and transmission (TX) using repeating bitmaps, each bitmap (a resource pool) is of certain size, say 20 (as per access configuration).

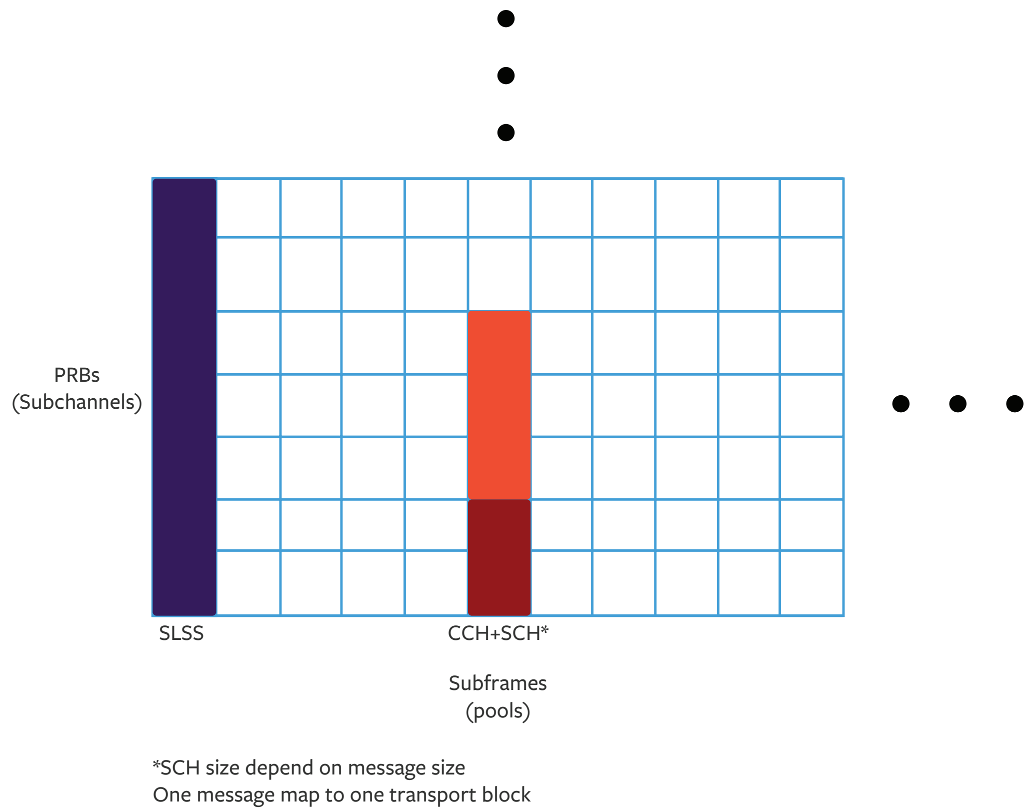


Fig 1: An example mapping of Sidelink physical channels

3.2 - Latency, Periodicity, and Message Size

Latency values are 20, 50, or 100 milliseconds. Periodicity is up to 10 messages per second (one per 100 ms). Message size could be as low as 50-300 bytes or as high as 1200 bytes.

In autonomous selection, one V2X message is transmitted over one subframe. So, in a way, **autonomous resource selection is basically a selection of a subframe** from available subframes at a particular location (or cell area).

3.3 - Vehicles in the Vicinity and Without 1-to-1 Communication

Most of targeted V2X scenarios require broadcast of the messages. So one-to-one communication is not needed. However autonomous selection does require sharing of certain (part of schedule) information between vehicles. This may be achieved via sidelink physical control channel (PSCCH). Sharing of full schedule is not right; vehicles change relative positions; for optimal operation, time between two transmissions can be broadcast but then number of transmissions over longer period (say more than a second) is not supposed to be considered in algorithm.

3.4 - Synchronization

A point which is not very apparent in problem statement and later is: How do vehicles know where sub-frame (say n) **start in time?** We shall need one vehicle to act as synchronization source or a common synchronization source.

3GPP allows two synchronization sources, LTE transmission itself or GNSS (satellites). It also says, where (LTE) network is not available, if a UE does not find V2V **synchronization signals**, it may synchronize with GNSS and then start transmitting periodic V2V **synchronization signals** (plus subframe number) at particular sub-frame timings. This shall act as synchronization source for other vehicles entering its vicinity.

3.5 - Access Configuration

Remember that access configuration is not only broadcast by network (V2X dedicated in SIB T21) but also available in UICC under V2X service tag. Of course, the configuration broadcast by network takes precedence when network is available.

Access configuration includes spectrum to use (band 47), bandwidth (10 or 20 MHz), number of PRBs per subchannel, sub-frame pools (RX, TX, and their availability bitmaps), and synchronization source (eNB or GNSS), retransmission (re-transmit once or do not), and few parameters that we shall encounter in next sections.

4 - Autonomous Selection

Now that we have covered background information and various points behind the problem and solutions, it is time to turn our attention to core algorithm of autonomous resource selection.

Set of rules have been devised, which if obeyed by all vehicles, shall result into (almost) collision free transmission or use of shared resources (subframes). Let us look at these rules.

4.1 - T1 and T2

Say V2X (ITS) application request a transmission of V2X message at nth subframe. Applications expect the message to be transmitted within latency period, say T2 ms. As one subframe is of 1 ms duration, it is as good as saying T2 subframes. System take certain finite amount of time to "handle or process" the message, let us call it T1 subframes (T1 <= 4, 4 is limit set by rules).

Effectively, algorithm has to select a right subframe from set {T1, T1+1, ... T2 }.

For the time being, we shall assume, any one subframe from this set can be chosen for the current transmission - *we shall loop back in here*.

4.2 - RRP

Say subframe N is randomly selected. Current V2X message is to be transmitted in Nth subframe. Later transmissions may (only) be made in N + RRP, N + 2*RRP, ... N + (slctr-1)*RRP subframes.

RRP stand for Resource Reservation Period. Value of RRP is to be chosen by UE implementation; valid values are part of access configuration from set {20, 50, 100, 200, ... 1000}. Value of slctr is to be randomly selected from certain range depending on RRP value (e.g. between 5 to 15 for RRP=100). Ranges are specified in MAC specification.

UE need to transmit RRP value in control information.

4.3 - *Looping back*

We now looped backed to selection of subframe from set {T1, T1+1, ... T2 }. For UE to correctly choose the subframe, it has to extrapolate from earlier receptions. Maximum resource reservation period is 1000ms, so reception data for last at least 1000 subframes are needed. If any such subframe y, reception had been received with RRPy, one (or more) of (y + i*RRPy) values where i = 1, 2 .... shall fall in the window of T1 to T2. In those subframes, as reception is expected, UE should eliminate them from set {T1, T1+1, ... T2 }.

Above steps complete the basic algorithm, but then few possibilities arise, as detailed in below section.

5 - Further scenarios that arise

a. No subframe left or too few subframes to choose from It might happen that no subframes or too few subframes remain in selection set. Specification say at least 20% of subframes should be present in selection set before random one may be selected from. To achieve this, UE should eliminate extrapolated subframes for which earlier receptions were of lower signal strength than certain threshold value; threshold value come from access configuration. Elimination to be continued with every step increasing threshold by 3 db - till at least 20% subframes are in selection set.

b. What to do for transmissions after (N + (slctr-1)*RRP) th subframe Well, toss a coin for heads! UE should select random probability; if it comes lower than "resource keep probability" from access configuration, UE may use N' + RRP, N' + 2*RRP, ... N' + slctrnew*RRP subframes for further transmissions (N' = N + (slctr-1)*RRP)). slctrnew is new random value. If probability do not favour, UE needs to redo the steps of extrapolation of subframes based on reception data of last 1000 subframes and so forth. It is easy to see that if "resource keep probability" is zero, it forces UEs to rerun algorithm after expiry of their autonomously selected schedule.

c. Transmission opportunities not used To ensure that autonomously selected transmission opportunities are utilised and not wasted beyond certain value, a rule is set. As per the rule, UE to stop using autonomously selected schedule after certain consecutive number of TX opportunities are not used. This value is part of access configuration.

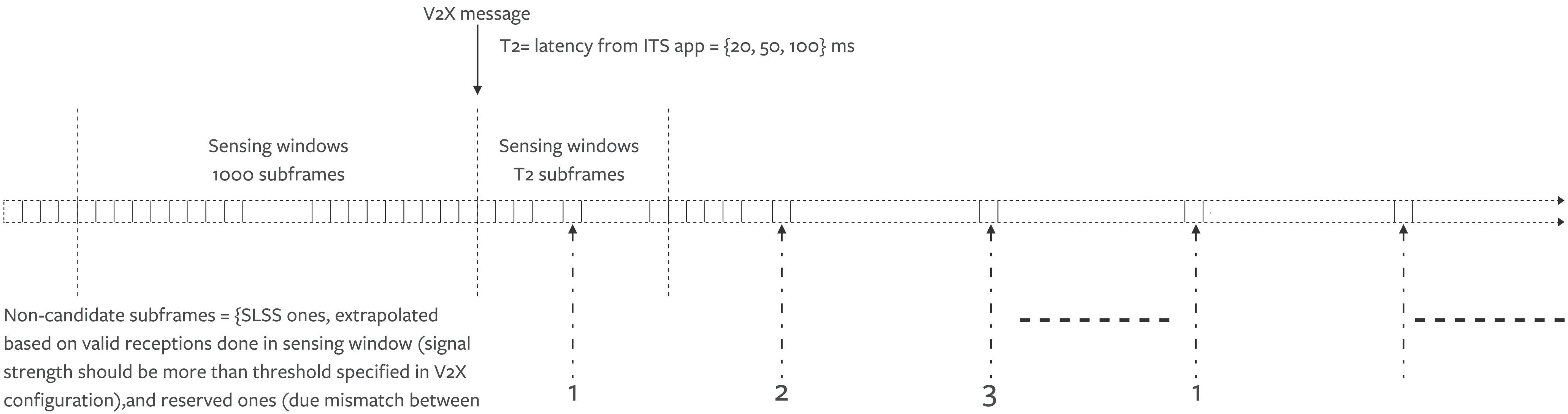
All the above cases are also shown in Fig 2.

6 - References

1. 3GPP TS 23.285 e60 "Architectural enhancements"
2. 3GPP TS 23.303 e10 "Proximity-based Services"
3. 3GPP TS 36.300 e60 "E-UTRAN Overall"
4. 3GPP TS 36.331 e62 "RRC"
5. 3GPP TS 36.321 e60 "MAC"
6. 3GPP TS 36.213 e60 "PHY procedures"
7. 3GPP TS 36.523-1 f10, section 24, "UE Conformance"
8. 3GPP TS 36.508 f10 "UE Conformance test environment"

   Patent WO2017176099A1

Remember bitmaps in RX and TX pool configuration mentioned in v2X configuration



Non-candidate subframes = {SLSS ones, extrapolated based on valid receptions done in sensing window (signal strength should be more than threshold specified in V2X configuration),and reserved ones (due mismatch between total and multiples of bitmap size)}.

Candidate subframes should be at least 20% of total possible subframes: if not, subframes with lowest signal strengths are re-considered as candidate subframes till 20% condition is fulfilled.

rand=Random number, between certain range, range is based on reservation interval, ranges are as specified in MAC specification.

If probResourcekeep do not allow to continue the allocation, above procedure is repeated to get fresh schedule.

Few transmission opportunities might not get used; if this number exceed than certain configured value from V2X configuration, schedule is dropped and above procedure need to be repeated when next time transmisiion is required.

Fig 2: CV2X Autonomous Resource Selection explained

C-V2X Autonomous Resource Selection Explained

marketing@sasken.com | www.sasken.com

USA | UK | FINLAND | GERMANY | JAPAN | INDIA

Jan 2019